

PARKSPHERE CONNECT

An IoT-based smart parking system that automates slot detection, gate control, and real-time cloud monitoring — designed to eliminate parking guesswork for good.

IOT SYSTEMS PROJECT

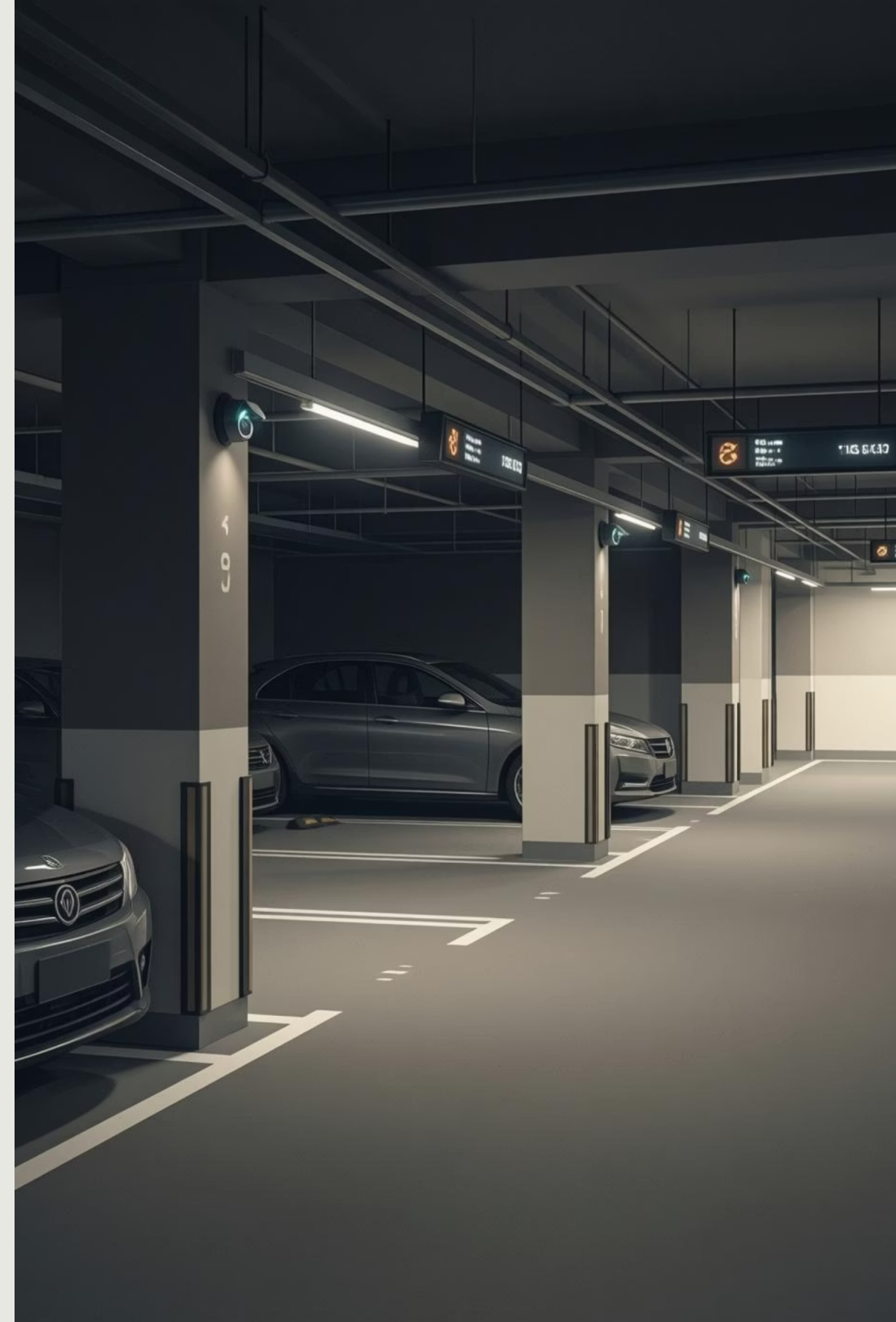
GROUP 7

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THE PROBLEM WE'RE SOLVING

URBAN PARKING TODAY

Drivers waste significant time circling for open spots — burning fuel, worsening traffic, and causing frustration in malls, offices, and university campuses alike.

WHAT'S MISSING

- No real-time slot availability data
- Manual or zero monitoring systems
- No automation at entry/exit gates
- No remote oversight for parking

PROJECT OBJECTIVES

1

AUTOMATE SLOT DETECTION

Use ultrasonic sensors to continuously detect vehicle presence and determine real-time parking availability without manual checks.

2

CONTROL GATE ACCESS

Drive a servo motor to open or close the barrier automatically based on live sensor data — only granting entry when a slot is free.

3

ENABLE REMOTE MONITORING

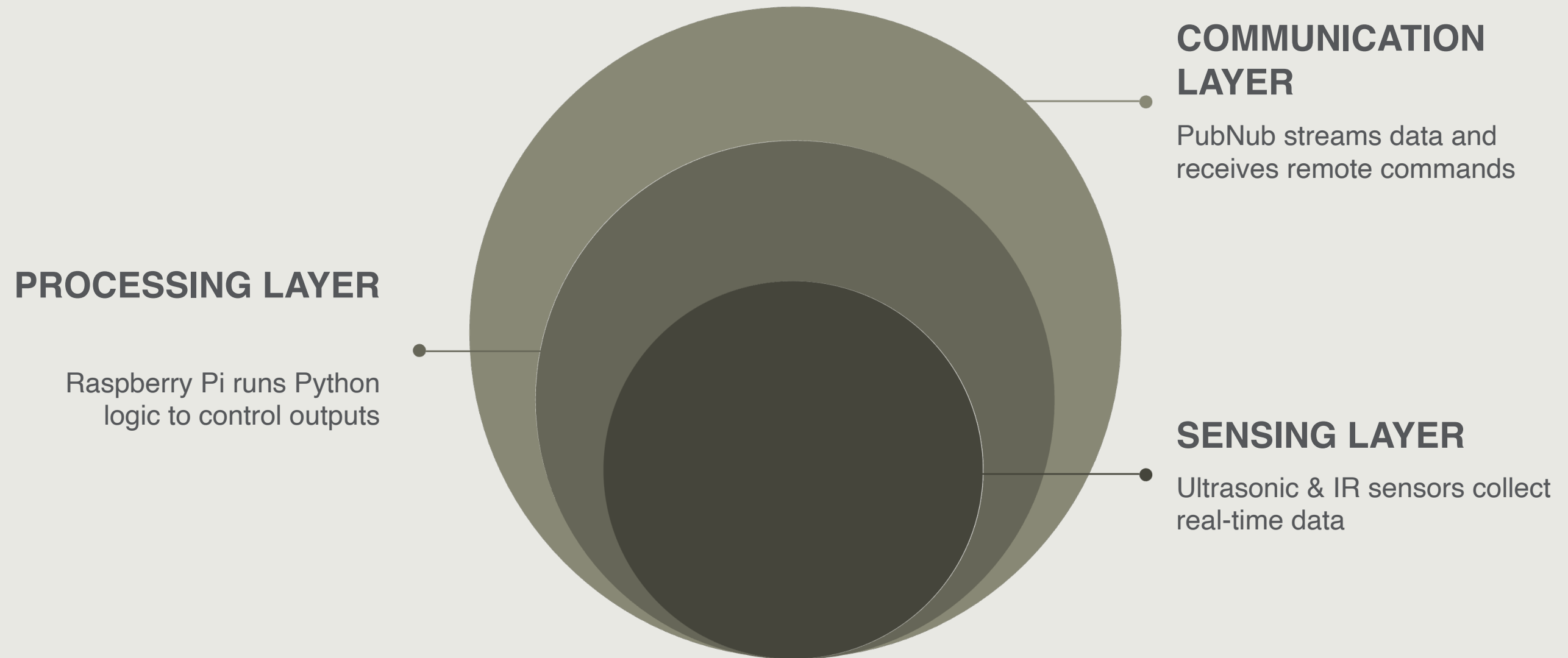
Integrate PubNub cloud communication so parking status, distance readings, and gate state are accessible from anywhere in real time.

4

ENSURE SAFETY & FEEDBACK

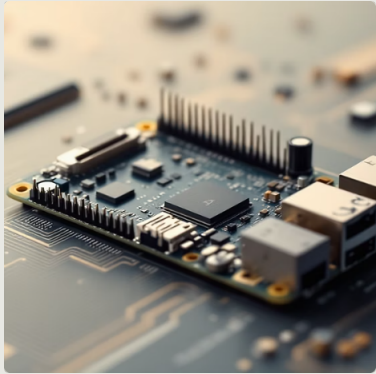
Provide LED visual cues and buzzer alerts to guide drivers safely, reducing blind-spot collisions and improving the overall user experience.

SYSTEM ARCHITECTURE: THREE LAYERS



Each layer has a distinct role — sensing, deciding, and communicating — working together as a seamless, closed-loop smart parking pipeline.

HARDWARE COMPONENTS



RASPBERRY PI 4

Central controller running Python — reads sensors, drives actuators, and handles cloud communication via Wi-Fi.



ULTRASONIC SENSOR

Measures distance using Time of Flight. Slot is "Occupied" if distance < 10 cm, "Free" if greater — reliable even in low light.



IR SENSOR

Detects vehicle presence at the gate entry via reflected infrared light — triggers the gate sequence instantly.



SERVO MOTOR + LED & BUZZER

Servo controls the physical barrier gate (PWM at 50Hz). LED shows slot status; buzzer warns when a vehicle is under 5 cm away.

SOFTWARE STACK

WHAT RUNS ON THE PI

PYTHON

Core logic: sensor reads, gate control, and cloud publish — all in a continuous loop using RPi.GPIO and the Time library.

PUBNUB SDK

Pub/Sub messaging: broadcasts JSON payloads (distance, slot status, gate state) to channel "group_7" every second.

WHAT USERS SEE

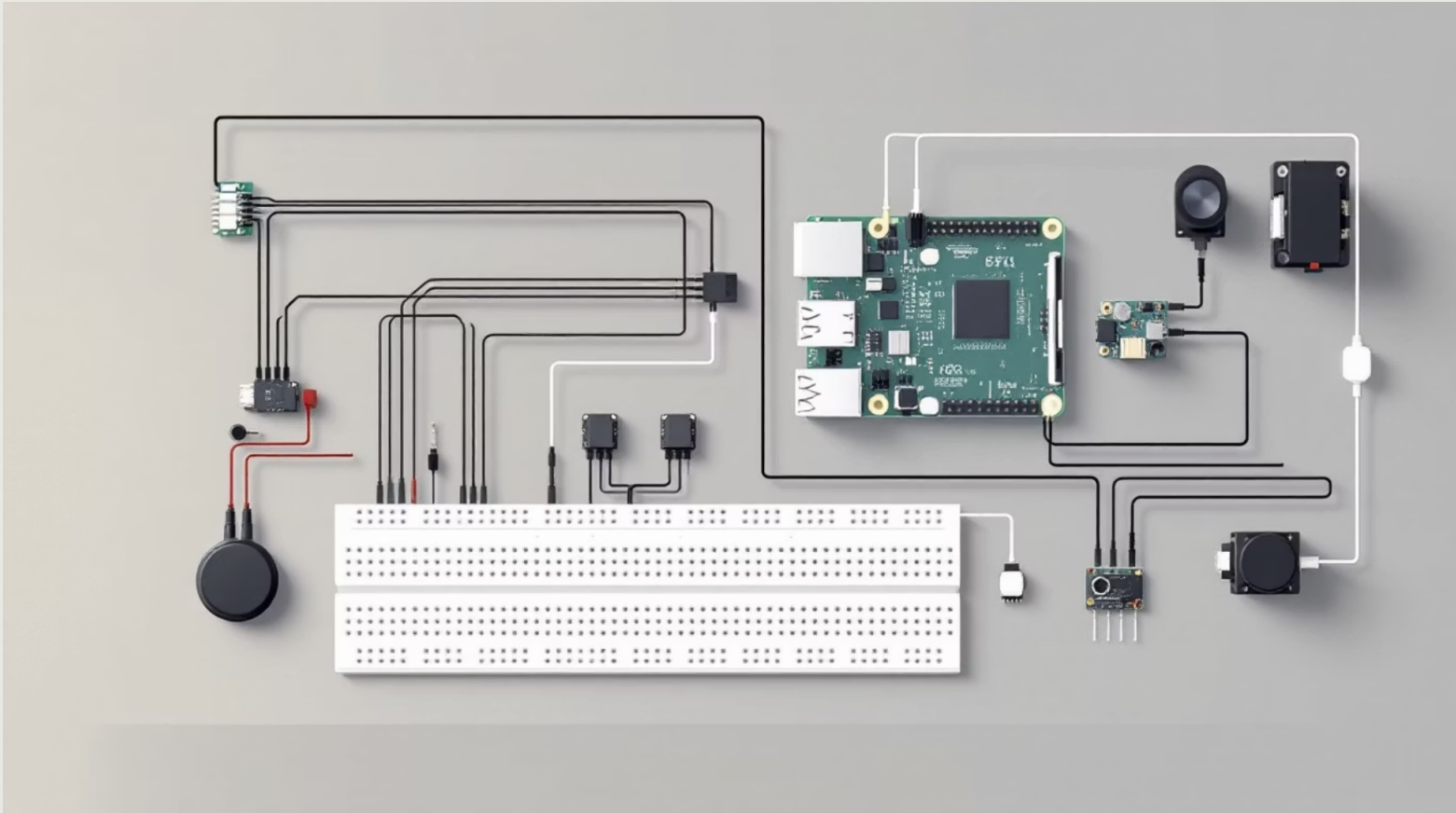
REACT.JS DASHBOARD

Subscribes to PubNub in real time. Displays a live parking map (green/red), exact sensor distance, gate state, and a full activity history log.

REMOTE OVERRIDE

Admins can send `force_open` or `force_close` commands from the web — the Pi listens and responds instantly via `SubscribeCallback`.

CIRCUIT & PIN MAPPING



GPIO PIN ASSIGNMENTS

Component	GPIO Pin	Type
Ultrasonic Trig	GPIO 23	Output
Ultrasonic Echo	GPIO 24	Input
IR Sensor OUT	GPIO 22	Input
Servo Motor	GPIO 12	PWM Out
LED Indicator	GPIO 18	Output
Buzzer	GPIO 20	Output

Gate opens at 7% duty cycle, closes at 2%. Buzzer triggers below 5 cm; LED lights when slot is free (>10 cm).

RESULTS: IT WORKS

100%

GATE ACCURACY

Servo responded correctly every time
IR + ultrasonic conditions were met
during testing

<1S

CLOUD LATENCY

PubNub delivered sensor updates to
the React dashboard in under one
second, consistently

4

FEEDBACK LAYERS

LED, buzzer, servo motor and cloud
dashboard all operating
simultaneously for full situational
awareness



LIMITATIONS & FUTURE SCOPE

CURRENT LIMITATIONS



Single-slot prototype — not yet scaled to large lots



Network dependency — cloud features fail without stable Wi-Fi



Sensor sensitivity — ultrasonic accuracy drops in dust, heat, or uneven surfaces



No mobile app — user interaction is limited to the web dashboard

WHERE THIS IS GOING



Multi-slot expansion with centralized dashboard for malls and campuses



Mobile app with navigation, booking, and payment integration



AI/ML predictions for parking availability based on usage patterns



License plate recognition and smart city infrastructure integration



KEY TAKEAWAYS

IOT WORKS IN THE REAL WORLD

ParkSphere Connect proves that low-cost hardware and cloud APIs can solve real urban challenges — no enterprise budget required.

SCALABLE BY DESIGN

The layered architecture — sensing, processing, communication — is built to grow from one slot to an entire smart city grid.

THE FOUNDATION IS SOLID

With AI prediction, mobile apps, and plate recognition on the roadmap, this prototype is just the beginning of a full smart parking ecosystem.

THANK YOU

We appreciate your attention and are now ready to answer any questions you may have about ParkSphere Connect.

GROUP 7

IOT SMART PARKING SYSTEM